

JAYASURYA BHASKAR

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SUMMARY

AI Engineer with **3+ years** building end-to-end AI software, with a focus on **computer vision**, **vision-language models**, and **LLM applications**. Experienced building scalable **AI pipelines** from data ingestion through production deployment, with hands-on ownership of company-wide **MLOps** and **release workflows**. Comfortable in fast-paced **R&D environments**, from rapid MVP prototyping to **production-ready** systems.

EXPERIENCE

AI Engineer <i>Castillo Engineering, Orlando, FL</i>	<i>May 2024 – Present</i>
- Built Castillo Planset QC, a production VLM pipeline using OpenAI/Gemini to review solar and electrical drawings against 470+ compliance checks; combined vision inspection with 18 deterministic NEC checks to reduce a 5-hour review to minutes. - Engineered adaptive tiling and coordinate remapping for large-format drawings using Gemini Vision; preserved small visual details and returned findings to precise plan coordinates, eliminating downsampling-related errors in the evaluation set. - Built PMO 360 with FastAPI, PostgreSQL, React, and OpenAI to turn meeting notes into structured actions and client deliverables; centralized project follow-up across PM teams with Microsoft Graph and Entra ID integration. - Trained a MobileNetV3 classifier with OpenCV preprocessing to identify solar-installation images and route them to Gemini Vision for compliance review across a 350+ report/month permit workflow. - Built Python/PySpark pipelines to clean, deduplicate, and standardize roughly 3 TB of raw solar-plan data; stored reusable structured datasets in Delta Lake for downstream ML and analytics. - Owned the company-wide codebase and release workflow, including PR reviews, GitHub administration, deployment documentation, and mentoring interns across production AI projects.	

Machine Learning Engineer <i>Genpact, Remote</i>	<i>Feb 2022 – Aug 2022</i>
- Built a YOLOv8 + U-Net pipeline on panoramic dental X-rays to localize teeth, segment periodontal bone, and measure bone loss; deployed the workflow across 5 dental clinics using PyTorch and OpenCV. - Trained a GAN-based enhancement model in PyTorch for noisy and low-contrast dental radiographs, producing cleaner inputs for downstream detection and segmentation. - Engineered multi-camera vehicle-damage pipelines across 100K+ images; automated dataset ingestion and versioning with PySpark/Airflow, then tracked experiments and promoted validated models through MLflow and CI/CD.	

PROJECTS

Castillo Planset QC	Demo Code
Reviews solar and electrical drawings against compliance checks using Gemini Vision, OpenAI, and Python . Uses tiled visual inference to preserve fine details and map detected issues back to precise plan coordinates .	
PMO 360	Demo Code
Converts meeting notes into structured actions and client-ready documents using OpenAI, FastAPI, PostgreSQL, and React . Integrates Microsoft Graph and Entra ID for calendar, identity, workflow automation, and PDF/Word/Excel generation.	
AI Photography Culling	
Clusters visually related photos using DINOv3 embeddings and ranks the strongest frames using focus and image-quality scoring. Uses cluster-level visual context with OpenCV and LLMs to select final images and generate context-aware captions.	

TECHNICAL SKILLS

Computer Vision	PyTorch, OpenCV, YOLOv8, MobileNetV3, U-Net, DINOv3, GANs, Object Detection, Semantic Segmentation, Data Augmentation, Tiled Inference
Machine Learning	scikit-learn, XGBoost, Transfer Learning, Ensemble Learning, Feature Engineering, Hyperparameter Optimization
MLOps	MLflow, Experiment Tracking, Dataset/Model Versioning, Regression Testing, Model Monitoring, Docker, GitHub Actions CI/CD, Model Deployment
Cloud	Azure Container Apps, Azure OpenAI, AWS SageMaker, S3, Bedrock, GCP Vertex AI
AI / Multimodal	OpenAI, Gemini, Anthropic Claude, Claude Code, Vision-Language Models, Multimodal Pipelines, RAG, Structured Outputs, Agentic Workflows
Software	Python, SQL, FastAPI, REST APIs, PostgreSQL, SQLAlchemy, React/TypeScript, Microsoft Graph, Entra ID

EDUCATION

M.S. Computer Science (GPA: 3.7/4.0) – University of Central Florida, 2024
<i>Graduate Teaching Assistant, Advanced AI: Neural Networks, Deep Learning, NLP, Reinforcement Learning</i>